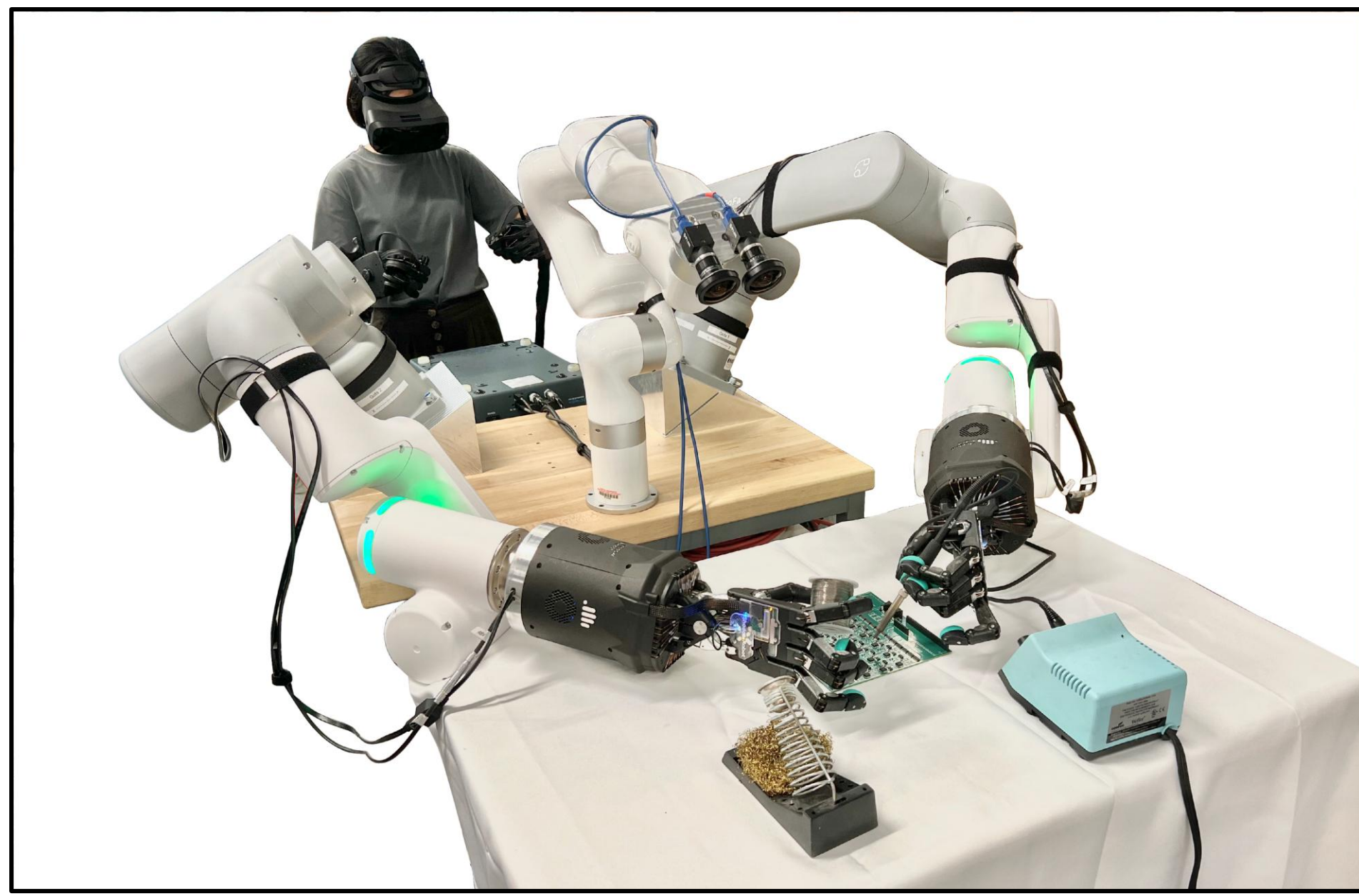


Sensorizing a Common Task to Characterize Grip Force Modulation and Finger Skin Deformation

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Motivation

- ❑ Current teleoperation platforms do not enable near-human performance during dexterous manipulation of objects.
- ❑ Sensorizing objects was shown to be insightful in characterizing how humans interact with objects under varying conditions.
- ❑ Previous works (e.g., [1]) have developed instrumented objects to measure manipulation forces and finger skin deformation.
- ❑ These instrumented objects only enabled characterization of lift-off tasks with single-finger imaging.



Teleoperation Platform (DexNex)

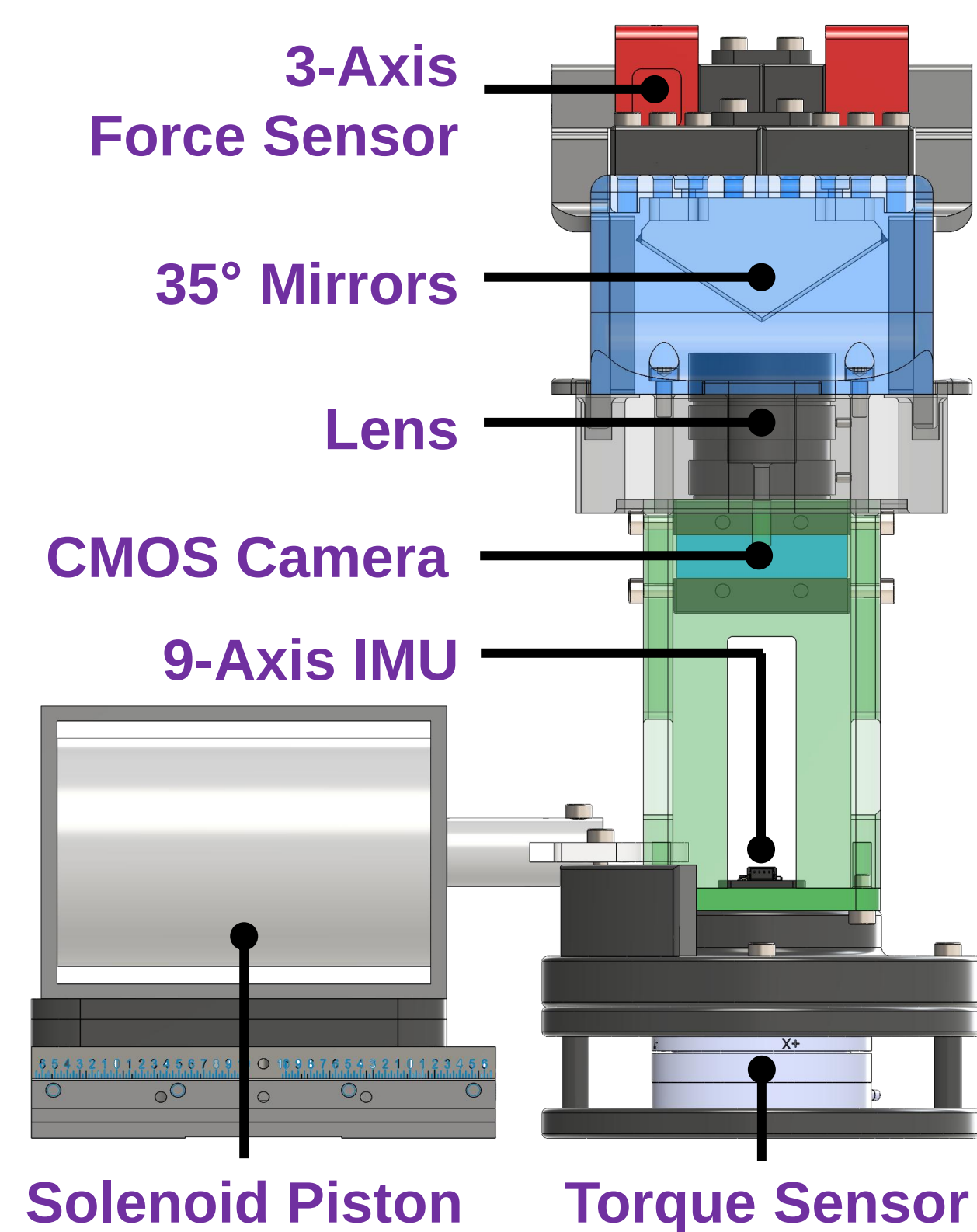


Instrumented Object [1]

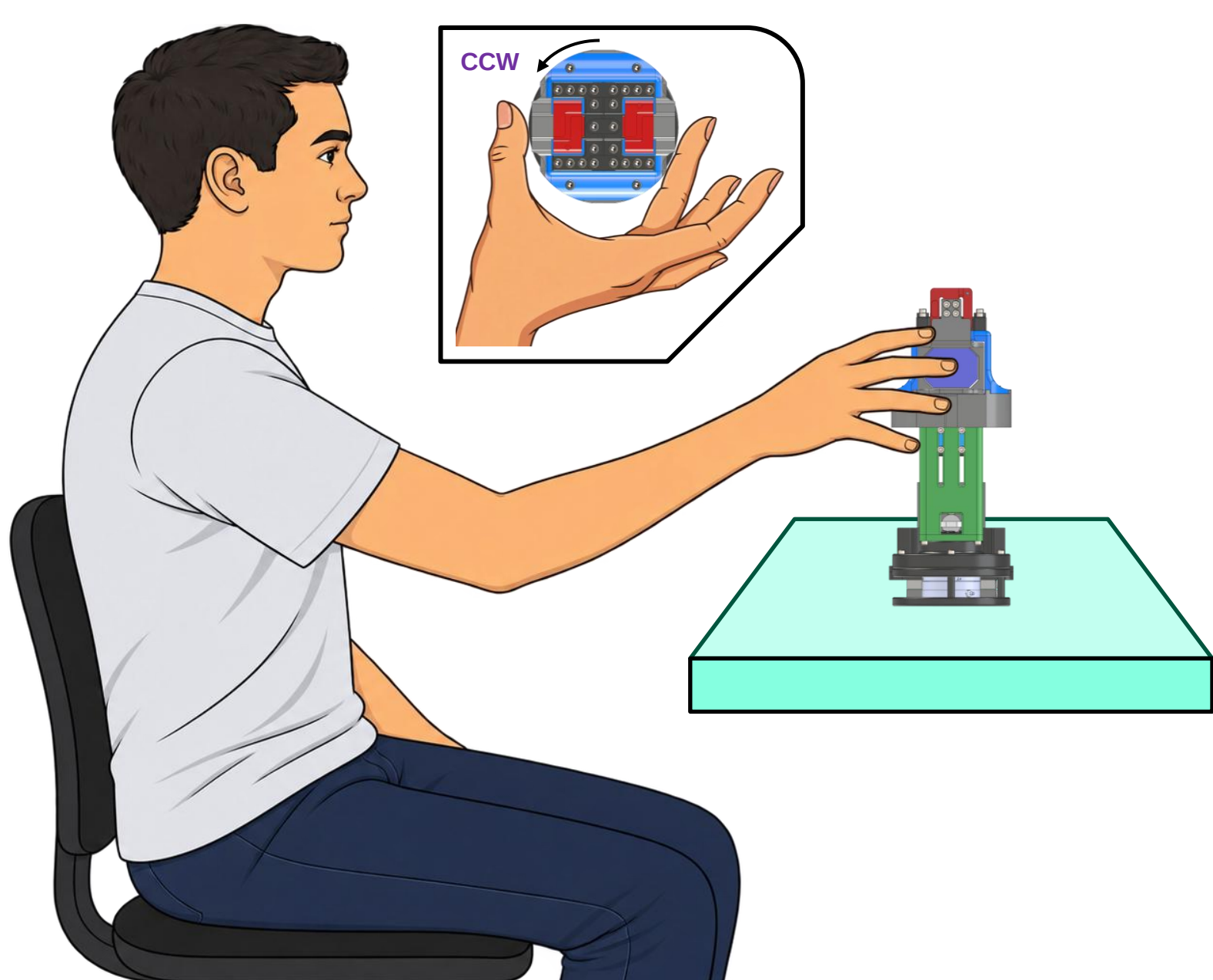
Can we characterize a more dexterous task while investigating finger-specific functional roles?

Methods

- ❑ We selected the task of opening a jar as the dexterous task to be investigated.
- ❑ We developed a jar-like instrumented object that mimics the mechanics of opening a real jar.
- ❑ The instrument enables per-finger normal and tangential force sensing and dual-fingerpad imaging.
- ❑ We use a lighting technique (Frustrated Total Internal Reflection) to illuminate the fingerpad contact area.
- ❑ We use optical flow algorithms to track fingerpad skin deformation during task execution.



Sensorized Platform



Jar Opening Trials

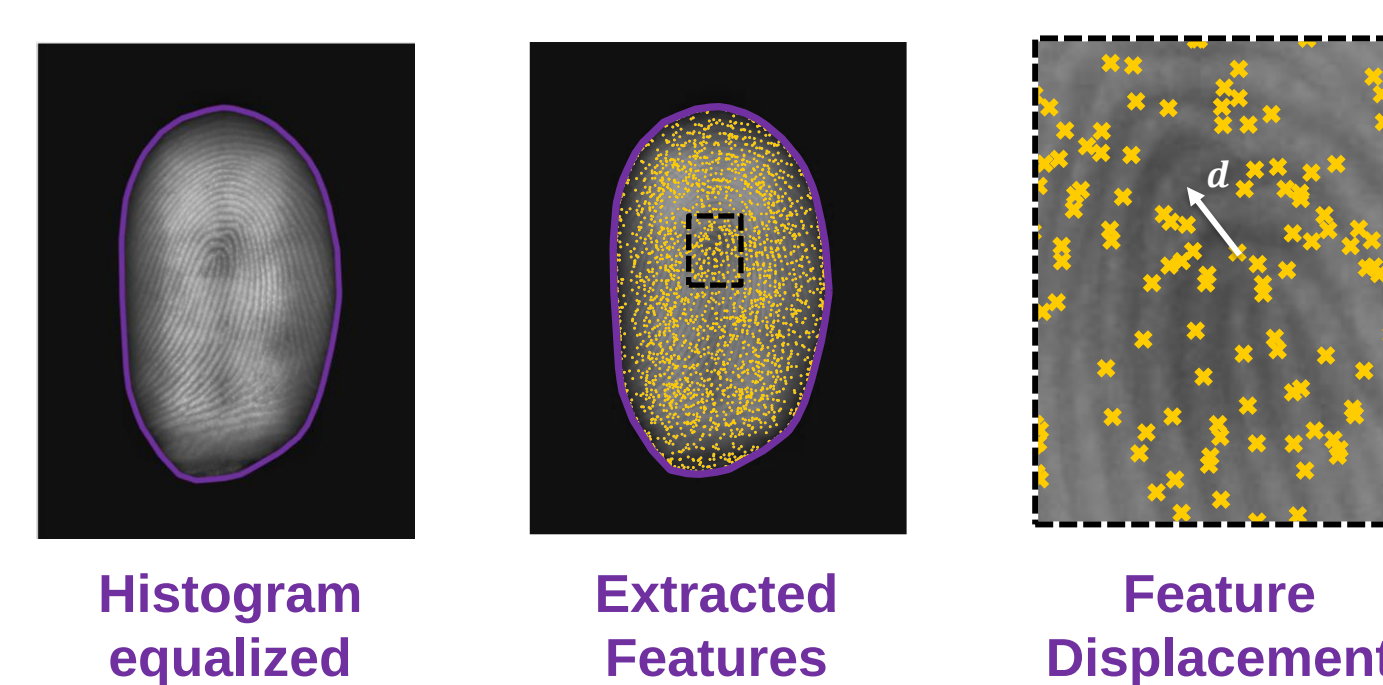
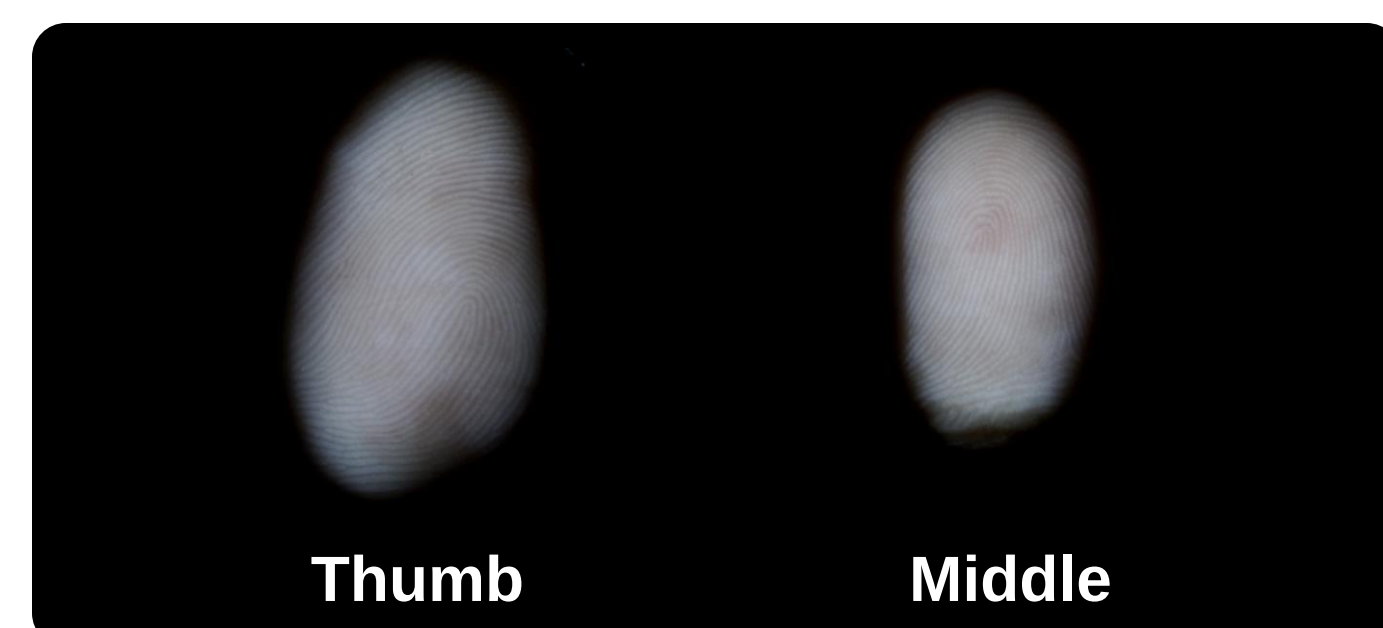
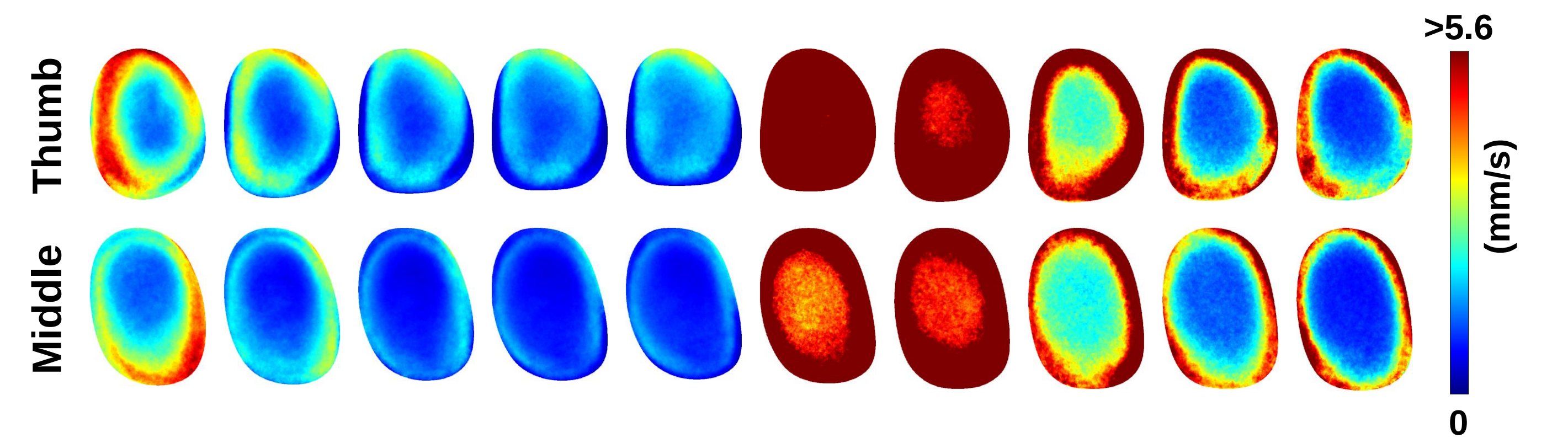


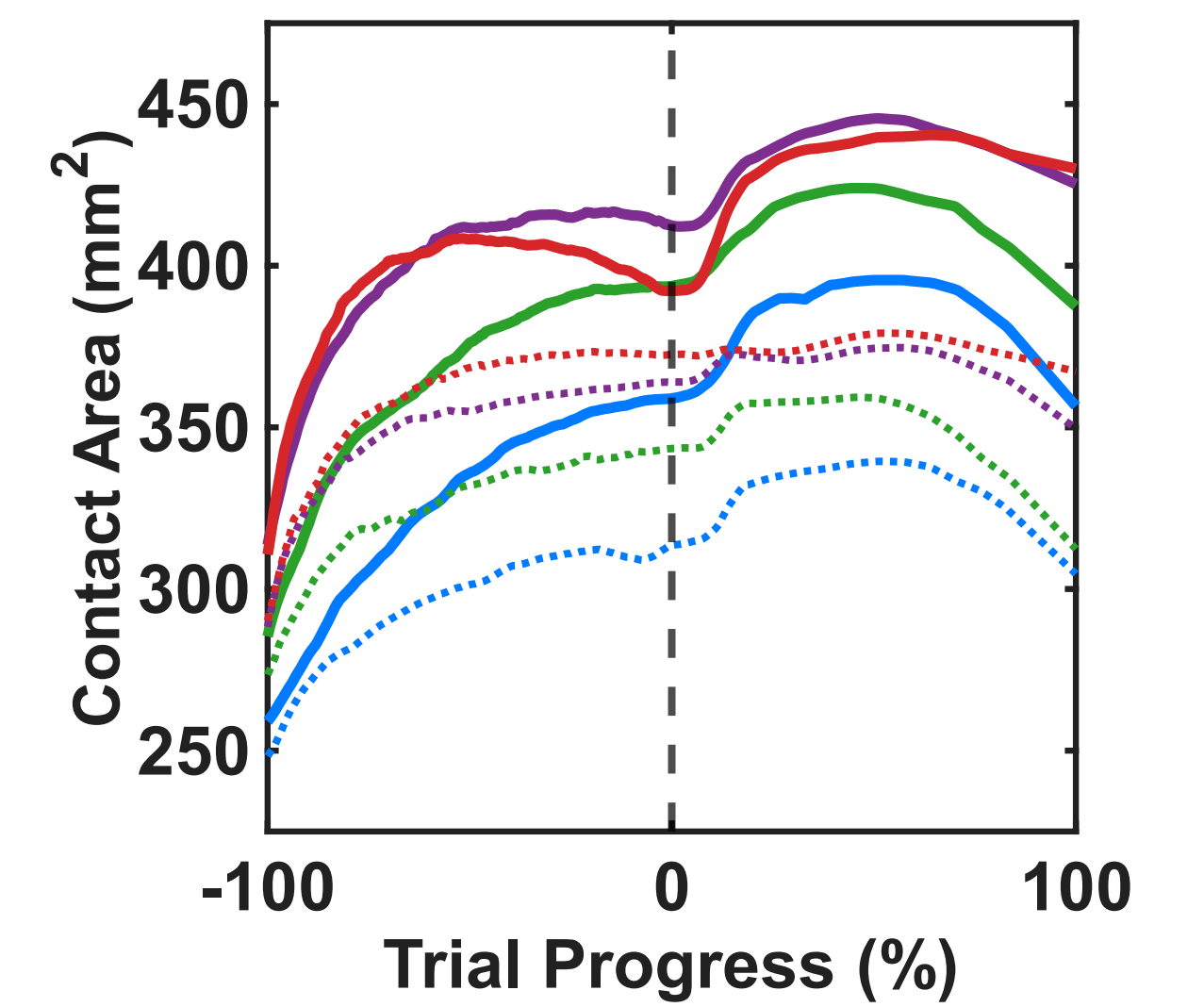
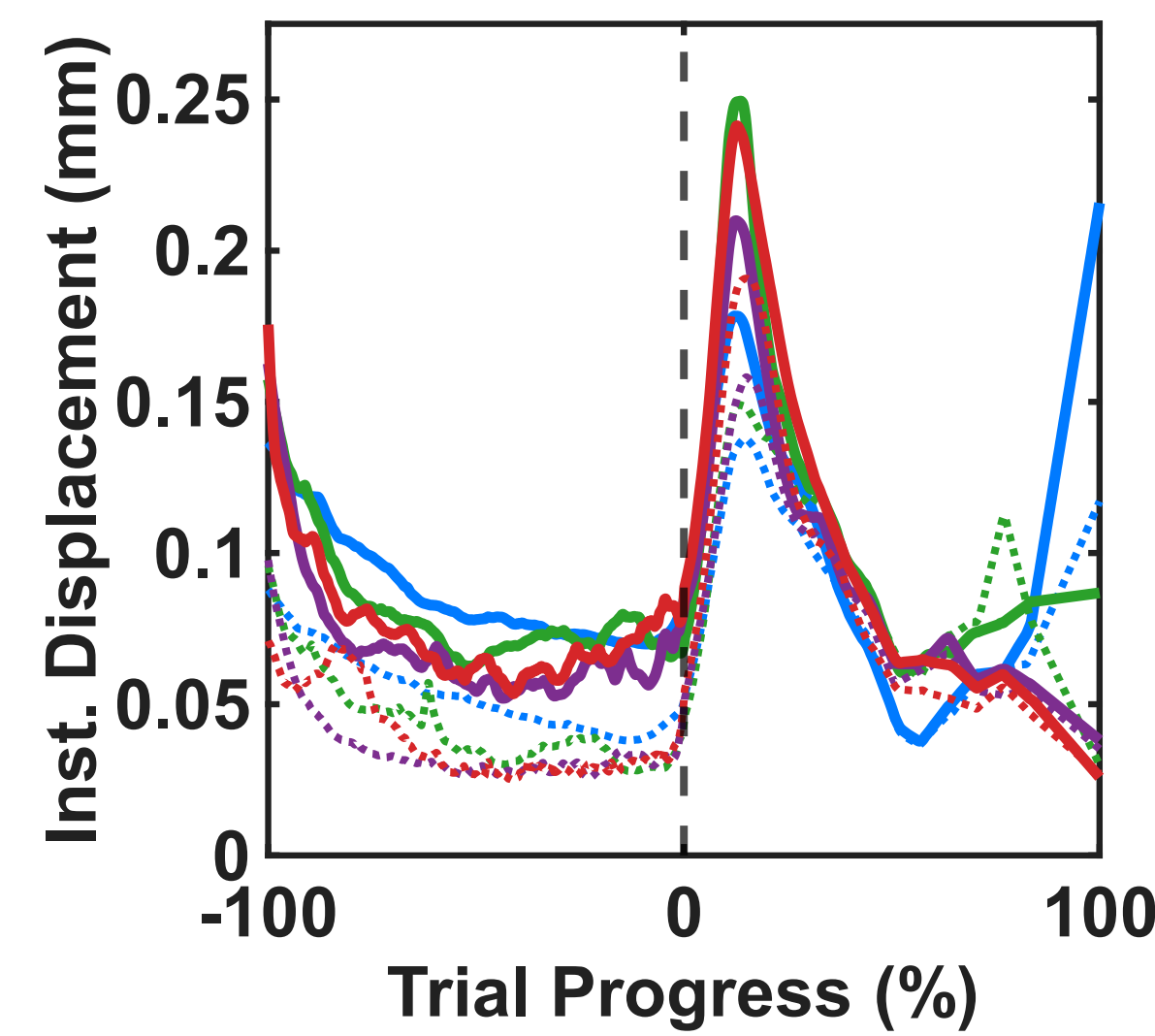
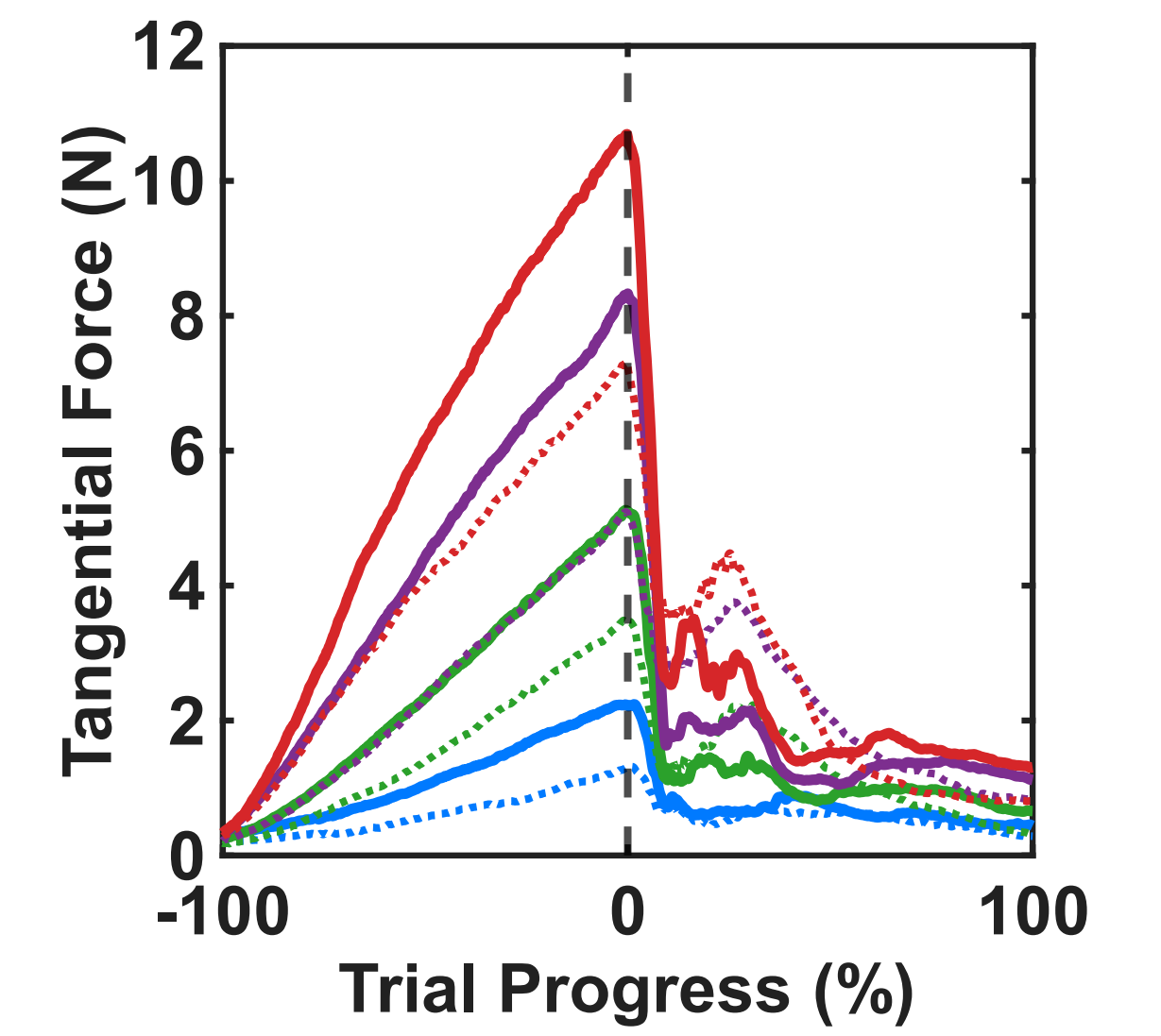
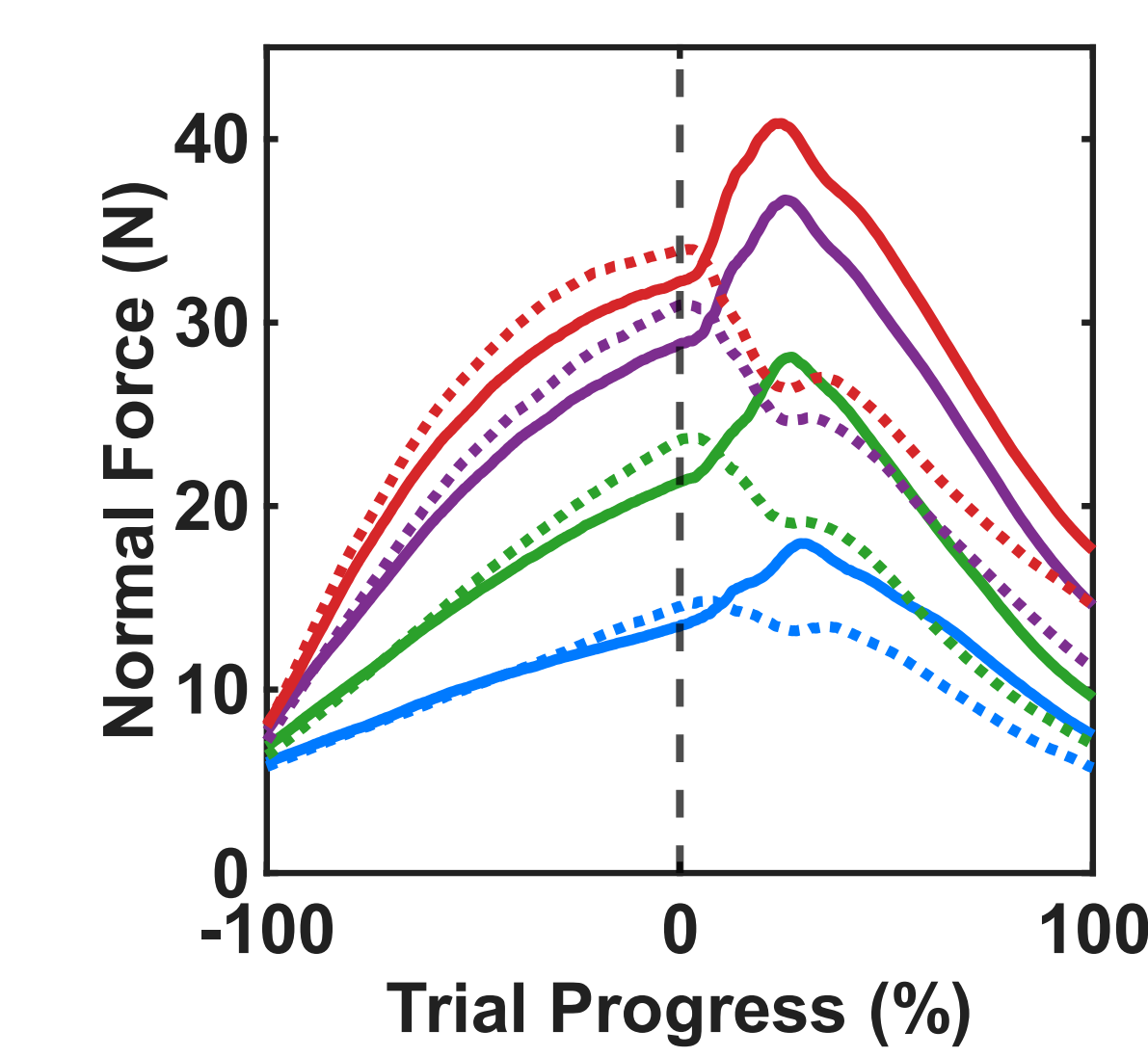
Image Processing Workflow

Results

We conducted a 16-participant human subject study to investigate functional independence between fingers involved in the task



— Thumb Middle — 0.5 N.m — 1.0 N.m — 1.5 N.m — 2.0 N.m



Thumb exerts lower Normal Force than **Middle**
Thumb exerts higher Tangential Force than **Middle**

Thumb exhibits higher Deformation than **Middle**
Thumb exhibits higher Contact Area change than **Middle**

Thumb likely played an actuating role while
Middle played a stabilizing role during jar opening

Despite changes in force distribution, similar displacements suggest that maintaining a preferred level of deformation may guide force distribution between the two fingers.

Future Work

We aim to utilize the concept of instrumented objects in investigating more dexterous tasks, as well as to use as a tool for benchmarking the performance of dexterous robots.

Acknowledgement

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References

- [1] Delhay, B. P., Schiltz, F., Barrea, A., Thonnard, J.-L., & Lefèvre, P. (2021). *Measuring fingerpad deformation during active object manipulation*. Journal of Neurophysiology, 126(4), 1455–1464. <https://doi.org/10.1152/jn.00358.2021>